Catarina Lucas Herrera

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EDUCATION

THE UNIVERSITY OF CHICAGO Odyssey Scholar, Humanities UX — Jun 2027
B.S. in Computer Science, Human-Computer Interaction; Masters Program in Computer Science
GPA: 3.52

MAJOR-RELATED COURSEWORK

Discrete Math, Theory of Algorithms, Database Systems, Intro. to Robotics, Intro. to Human-Computer Interaction, HCI: Engineering Interactive Devices, Systems with User-Centered Design, Actuated User Interfaces, Math Foundations for ML
MPCS Courses: Software Testing, Web Development, Design Build & Ship

SKILLS

- **Languages:** Proficient in reading and writing in English and Spanish.
- **Programming Languages:** SQL, C, C#, Python, Pytest, JavaScript, TypeScript, CSS, HTML, Claude Code, Github, Linux.
- **Frameworks & Practices:** React, Node.js, Next.js, Figma, Unity3D, Blender, Canva Adobe, CI/CD, OOP, Agile, Full-Stack

PROJECTS

VR Research Developer | UChicago Cinema and Media Studies — Jan - Aug 2026

- Programmed immersive VR environments in **Unity** to translate interdisciplinary research on illness, disability, and neurodivergence into interactive, embodied narrative experiences. Modeled, textured, and optimized 3D assets and environments in **Blender** for real-time VR rendering and custom **C#** script writing for unique interactions.

focusTACO | Creating Systems with User-Centered Design — Jan - Mar 2026

- Investigated, prototyped, and delivered a final app with a physical addressing dynamic student study habits and task management. Designed **Figma** buttons, pages, marketing design, hardware design, group **proposals**, and video editing.

Turtlebot4 "art-bot" | Introduction to Robotics — Dec 2025

- Successfully implemented an inverse kinematics algorithm using **object-oriented programming** in **Python** to translate cartesian coordinates into joint angles via **trigonometry** for a 3DOF robot arm. Incorporated a pre-built sign language detection model to translate letters from live camera feed onto a whiteboard using **computer vision.**

PROFESSIONAL EXPERIENCE

Artist's Books Exhibition — Chicago, IL
Jeff Metcalf Web Development and Management Intern — Oct 2026 – Jun 2027

- Actively developing a website E2E to host and display an artworks database in collaboration with Art History Department.

Bright Bots Initiative — Chicago, IL
Jeff Metcalf User Experience Lead Intern — Jun – Aug 2026

- Designed and built responsive, web-based educational mini-games and simulations in **React, TypeScript,** hosted on **Docker.**
- Applied UI/UX design principles: age-appropriate visual hierarchy, intuitive navigation, and clear instructional flows to create engaging interfaces for learners across K–2, 3–5, and 6–8 age bands following internationalization and accessibility guidelines.

University of Chicago Communications Rotational Internship Program — Chicago, IL
Jeff Metcalf Communications & Marketing Intern — Jan– Aug 2025

- Managed UChicago news and events for 1000+ users on **Sitecore & Asana**, part of the Internal Communications team.
- Storyboarded and edited videos, flyers, and captions, increasing engagement by over 200% on social media platforms.
- Researched, drafted, and published a landscaping story at UChicago, boosting engagement by 67%.

Code Your Dreams — Chicago, IL
Jeff Metcalf Computer Science Education Team Intern — Jun – Aug 2024

- Developed a CSS curriculum for the Edovo program devoted to educating incarcerated individuals across 750+ facilities.
- Redesigned the Code Your Dreams website by updating 10+ pages with organization data, appealing to website users.

LEADERSHIP & ACTIVITIES

Housing and Residence Life — Chicago, IL
Academic Year Resident Assistant, Summer Residential Assistant — Jun 2025 – Jun 2027

- Collaborated with professional staff to plan, promote and host study breaks with over 50% weekly resident participation.

University of Chicago Model United Nations (ChoMUN) — Chicago, IL
Bilingual Crisis Director (CD) — May 2025 – Apr 2026

- Collaborated with CDs to develop crisis arcs, background guides, and character briefs influenced by historical events.